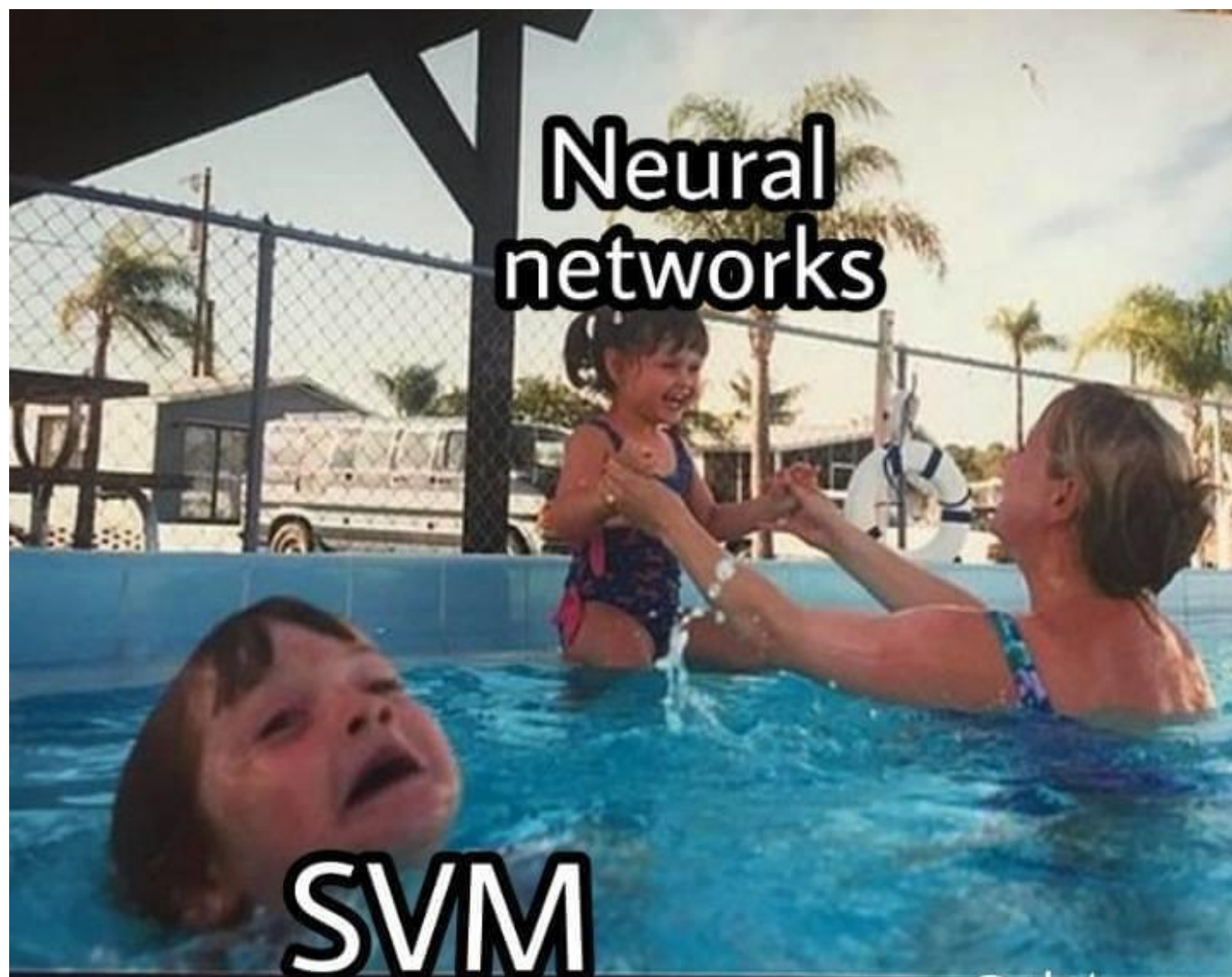


Neural Networks

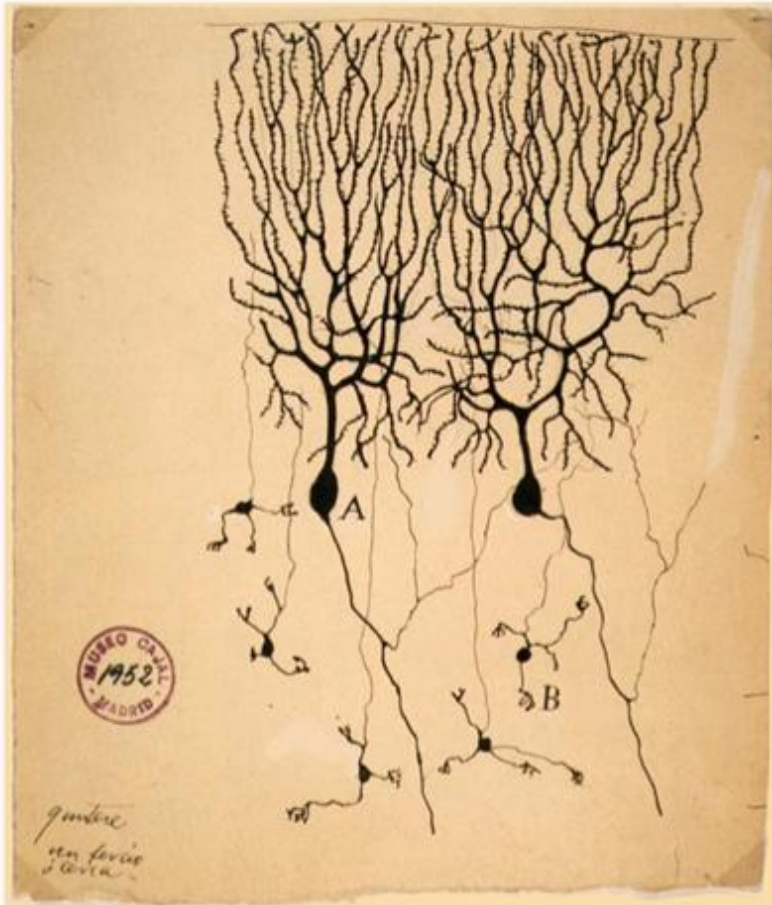
Forward Pass and Back Propagation

Mahdi Roozbahani

Georgia Tech



Inspiration from Biological Neurons

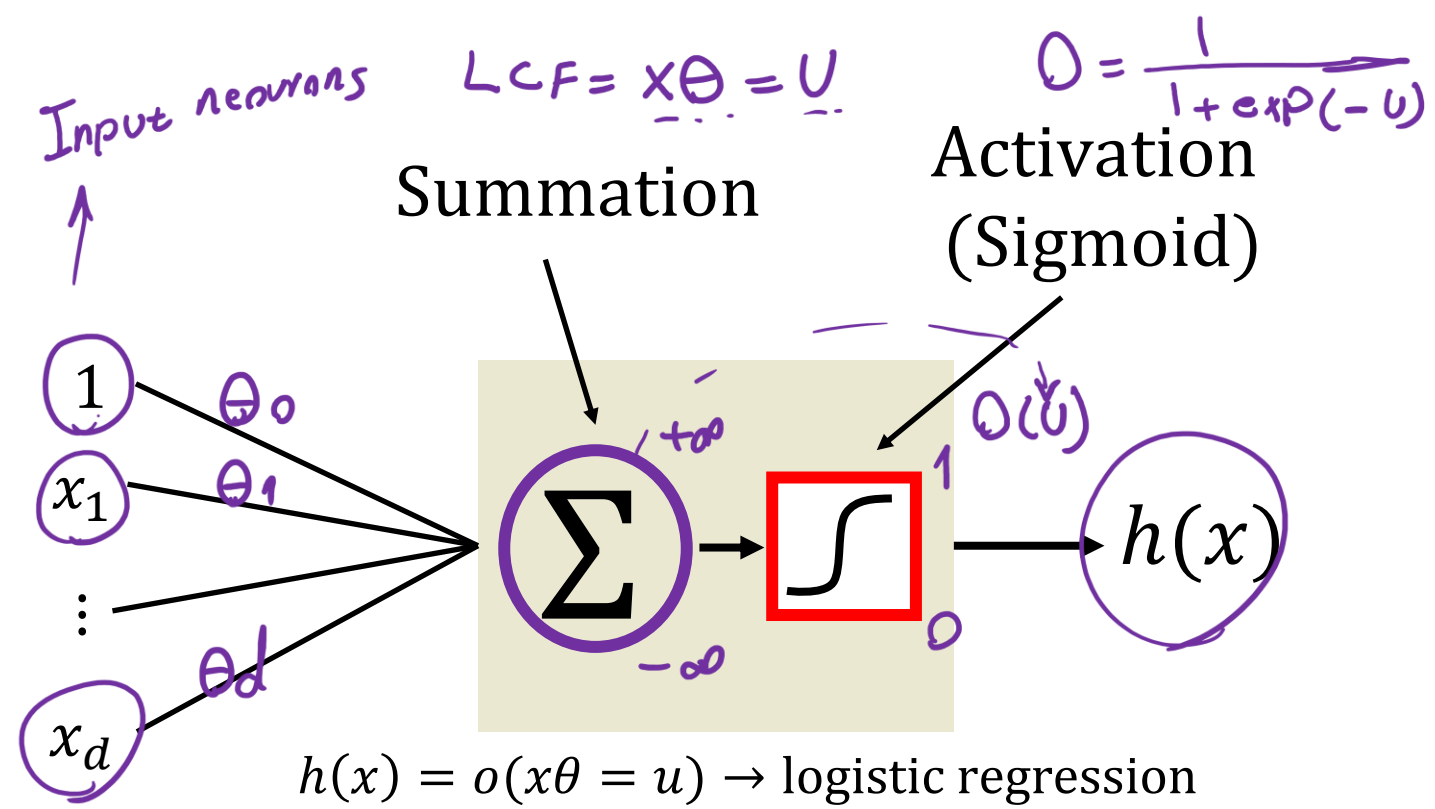


The first drawing of a brain cells by Santiago Ramón y Cajal in 1899

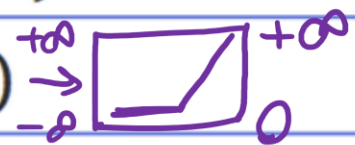
Neurons: core components of brain and the nervous system consisting of

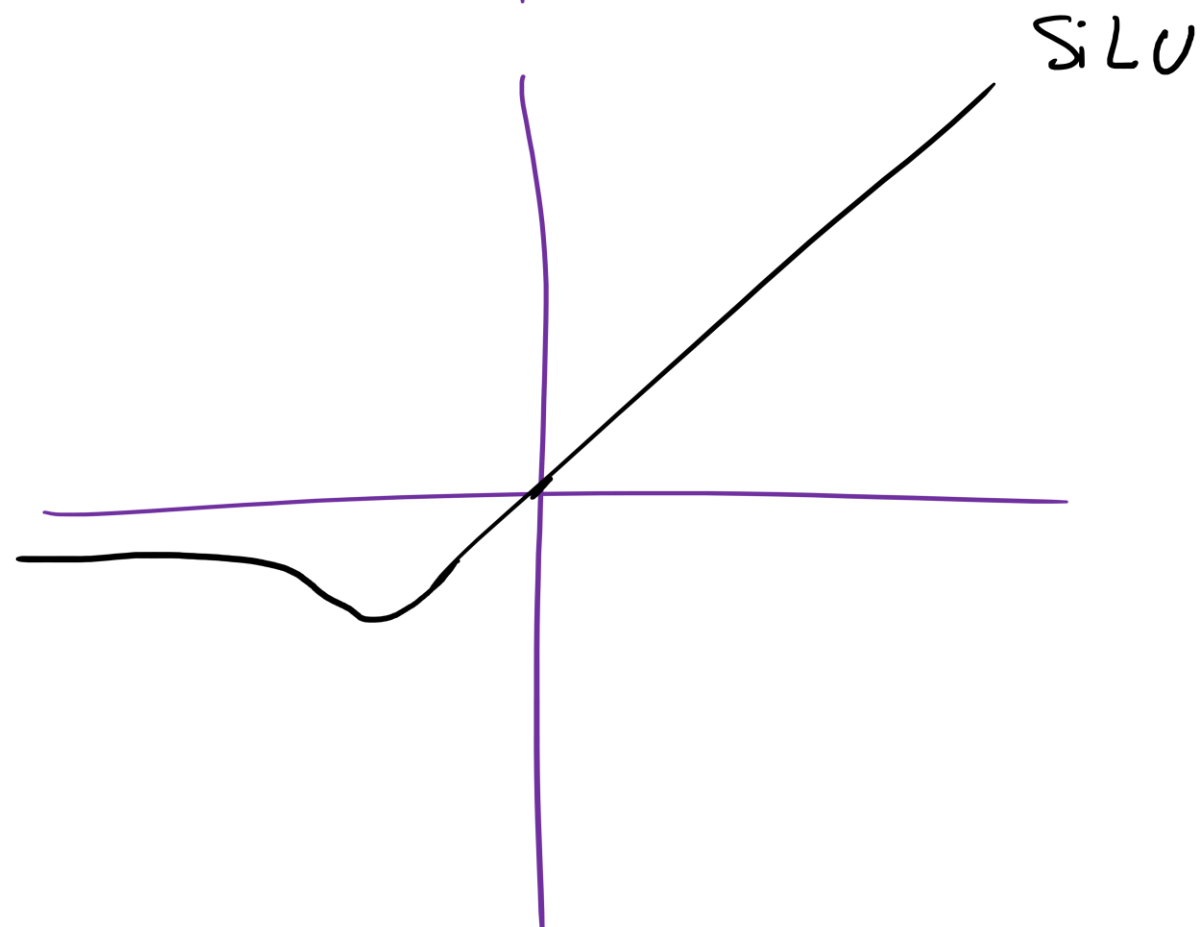
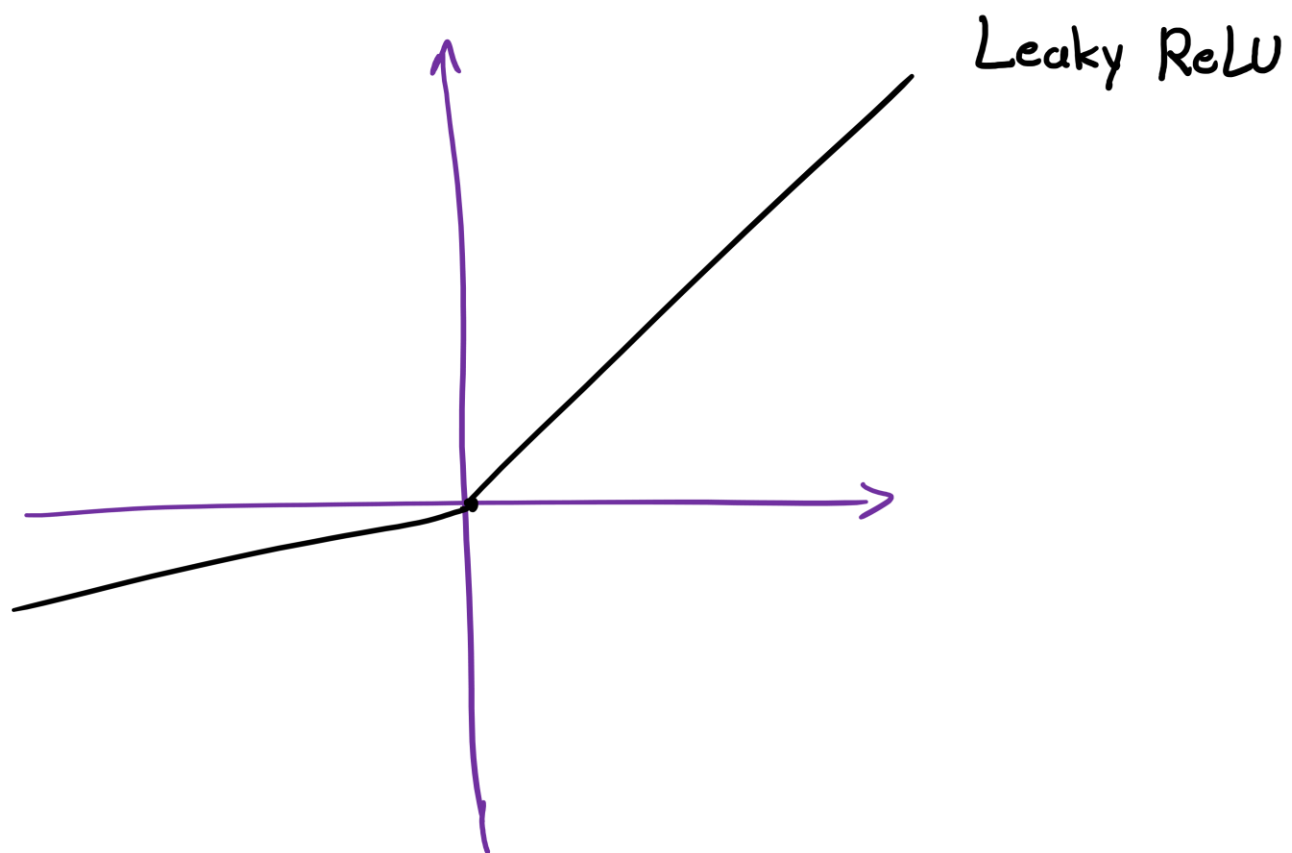
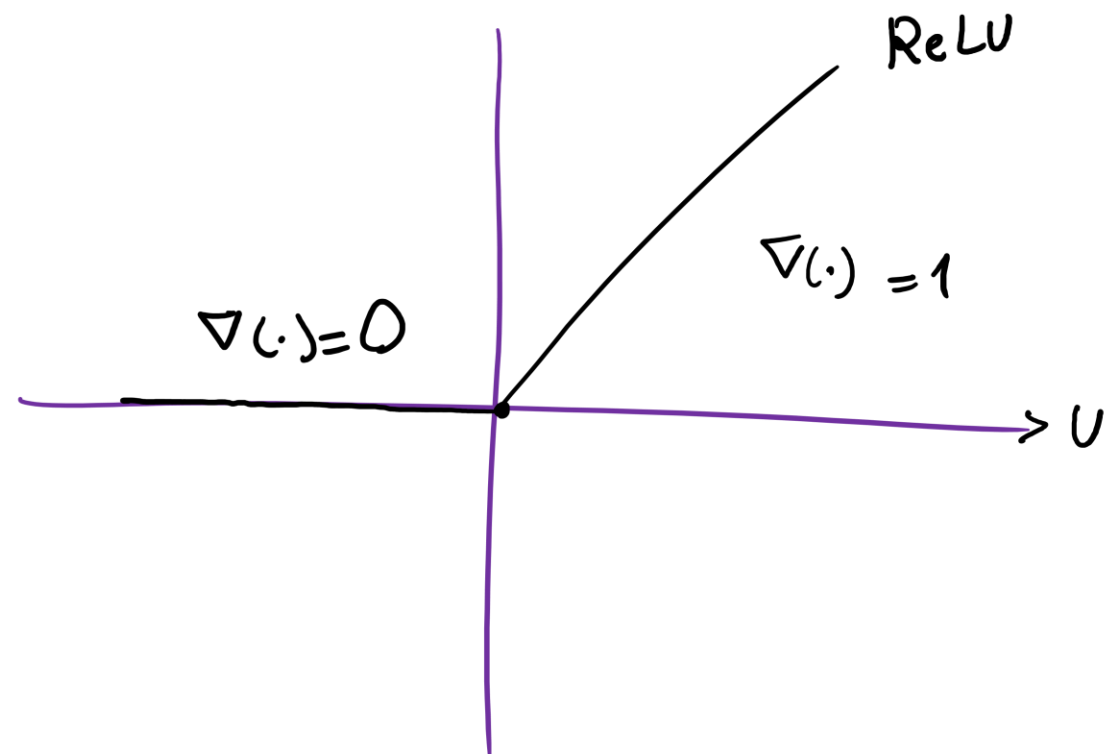
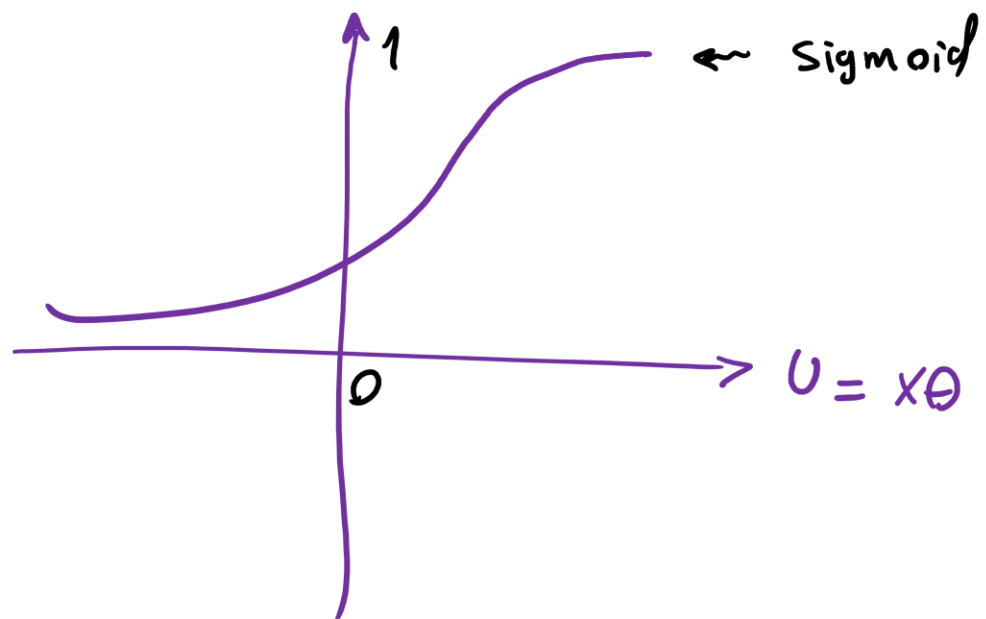
1. Dendrites that collect information from other neurons
2. An axon that generates outgoing spikes

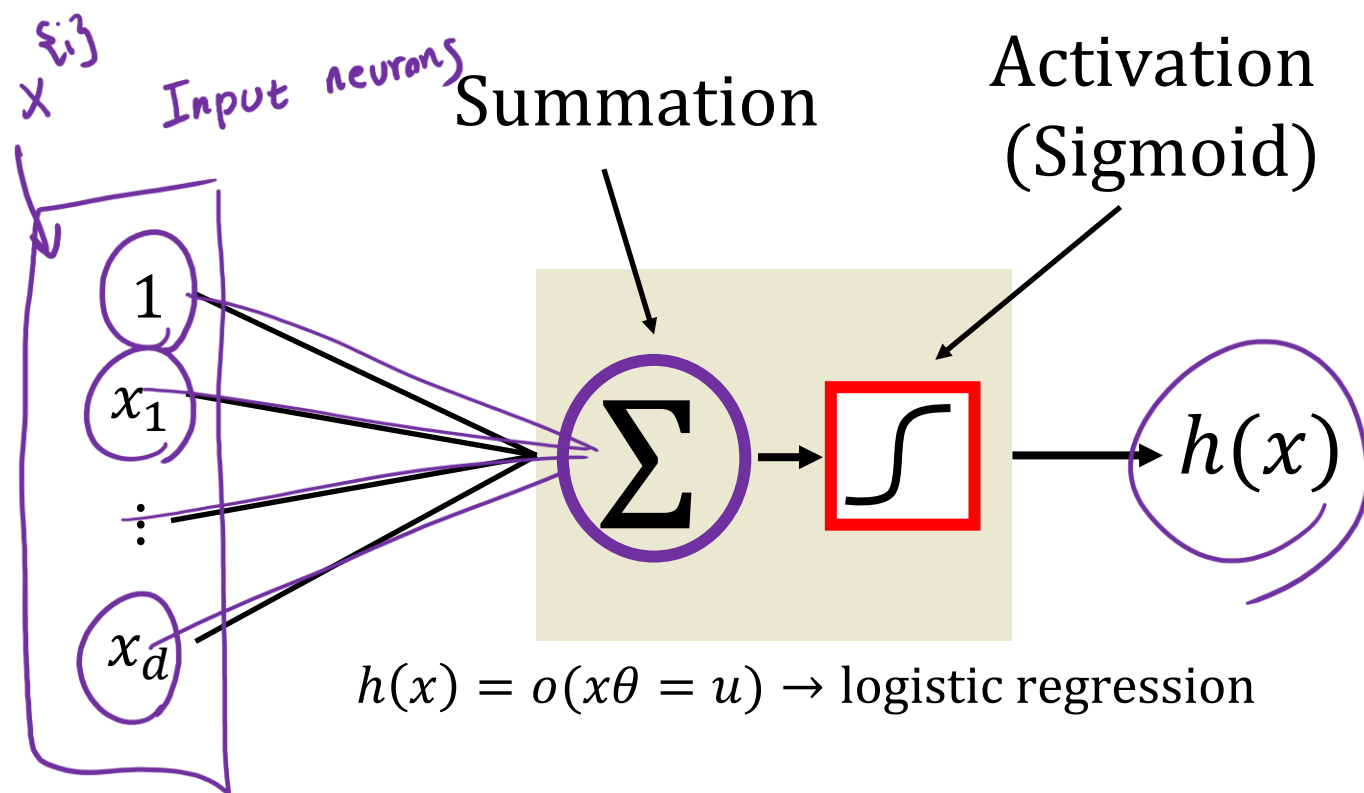
x



$$\text{output} = \text{activation}(x\theta + b)$$

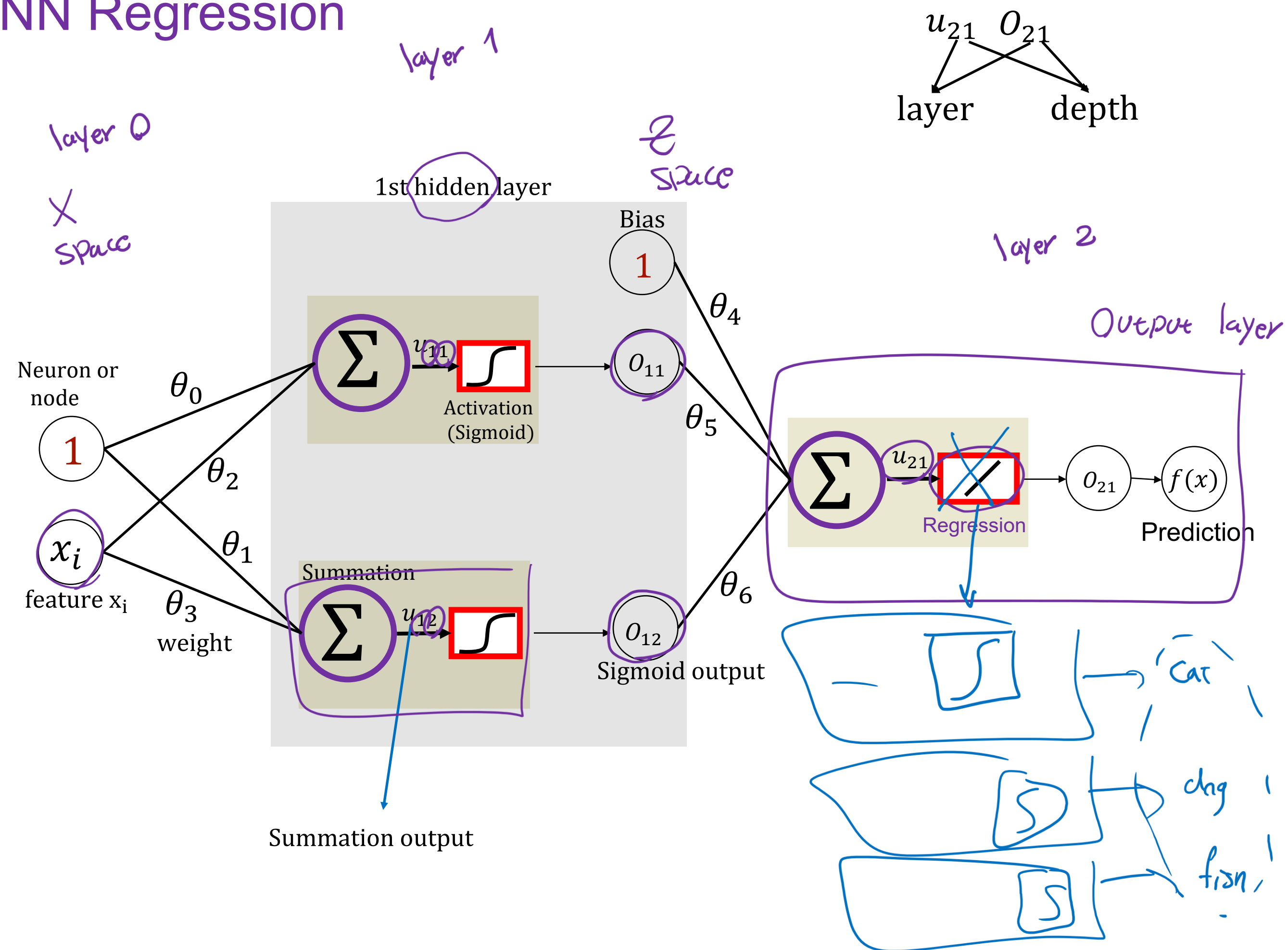
Name of the neuron	Activation function: $\text{activation}(z)$
Linear unit	$z \rightarrow$ 
Threshold/sign unit	$\text{sgn}(z)$ 
Sigmoid unit	$\frac{1}{1 + \exp(-z)}$ 
Rectified linear unit (ReLU)	$\max(0, z)$ 
Tanh unit	$\tanh(z)$ 



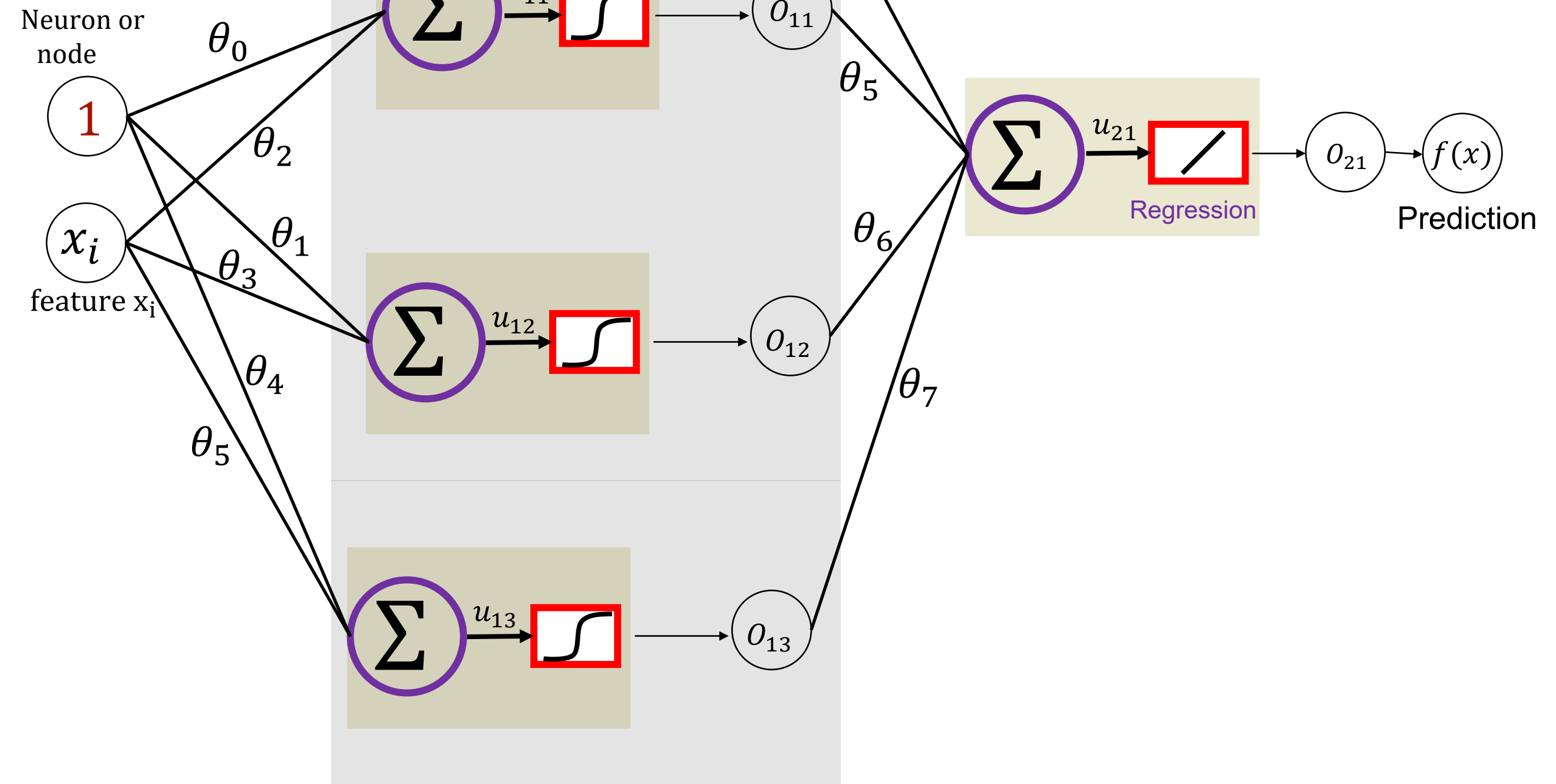


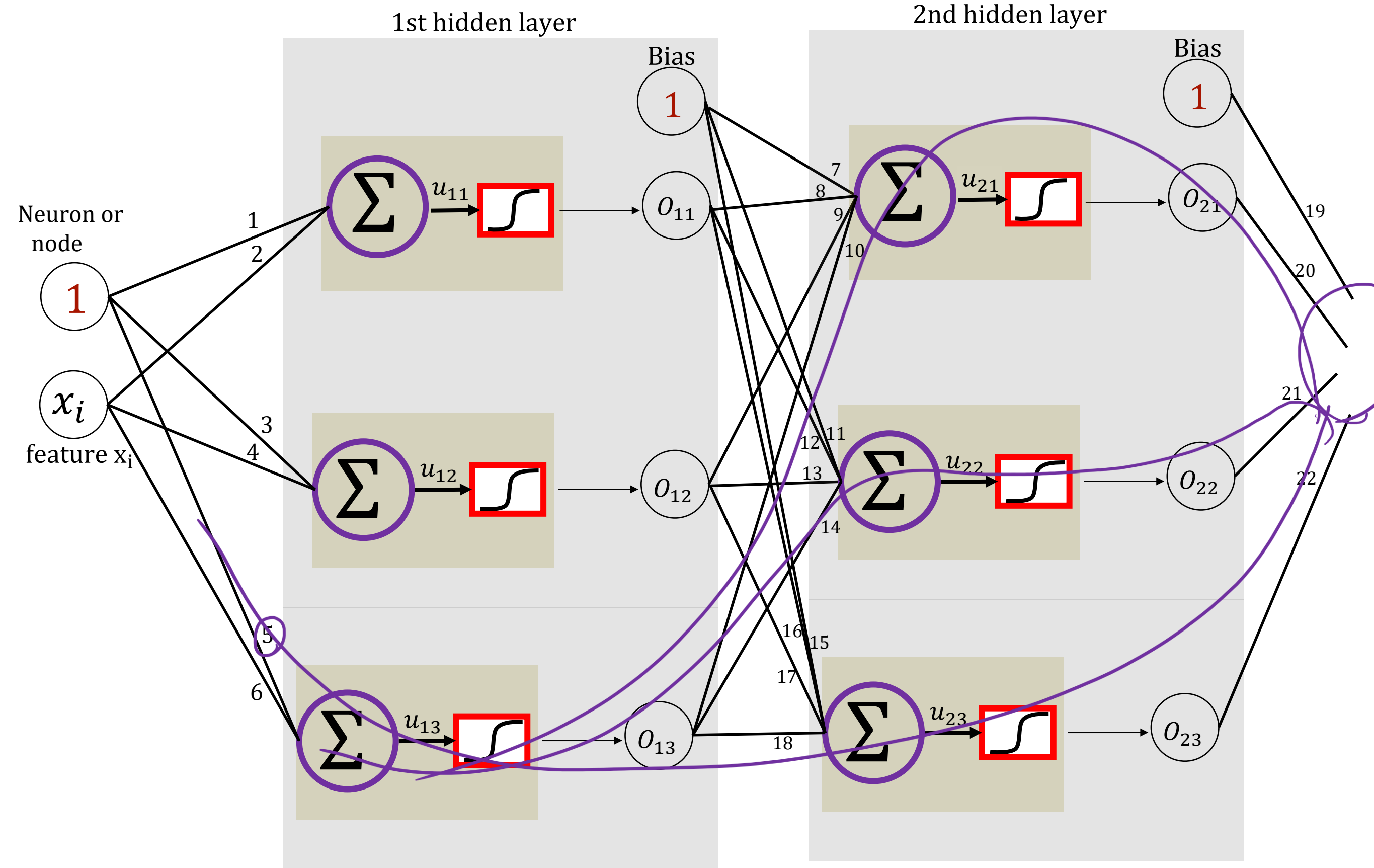
$\leadsto ANN \leadsto FC \text{ layer (Fully Connected)}$

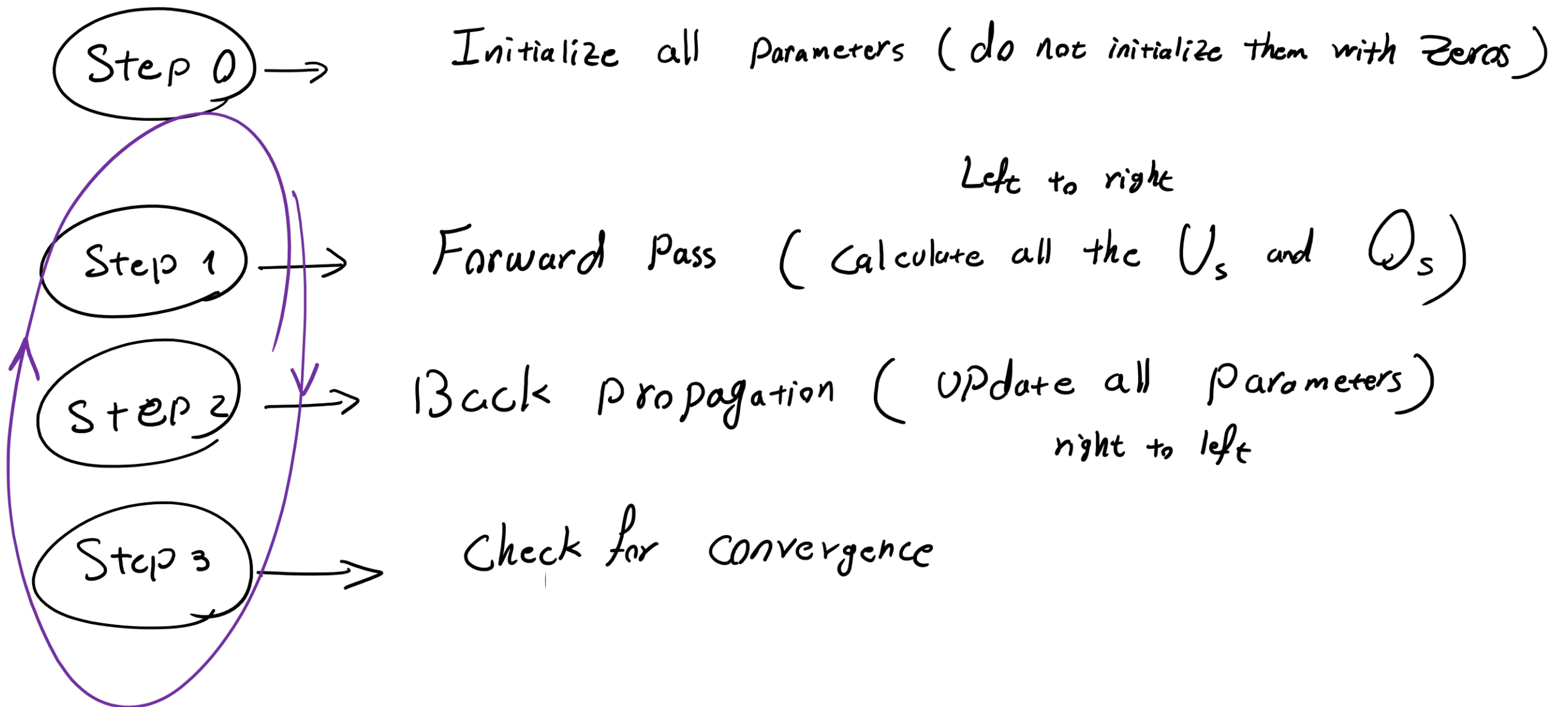
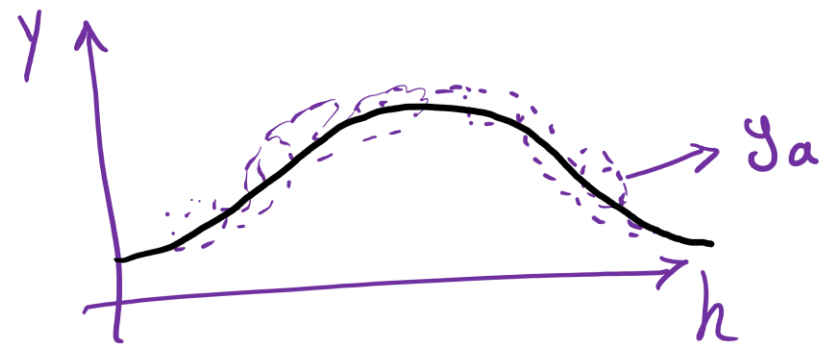
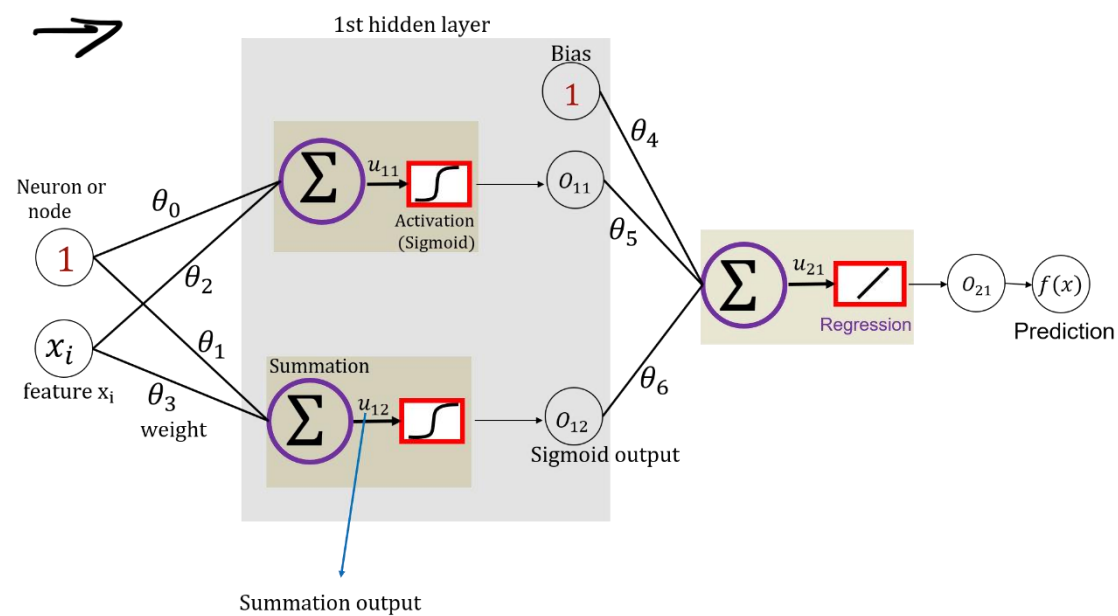
NN Regression

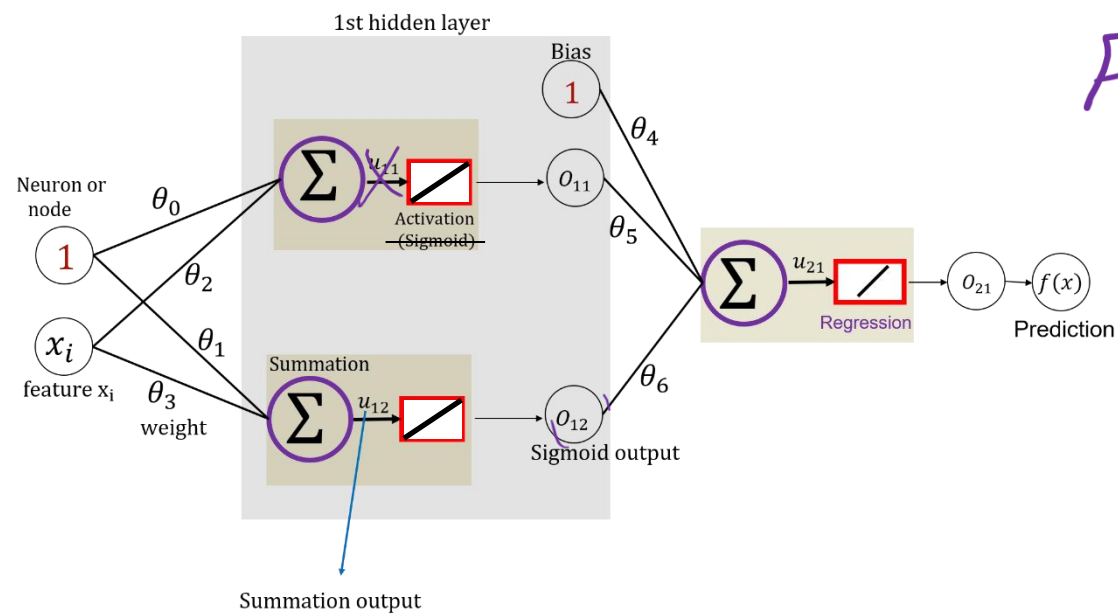


1st hidden layer









Forward pass



$$u_{11} = \theta_0 + \theta_2 x_i = o_{11}$$

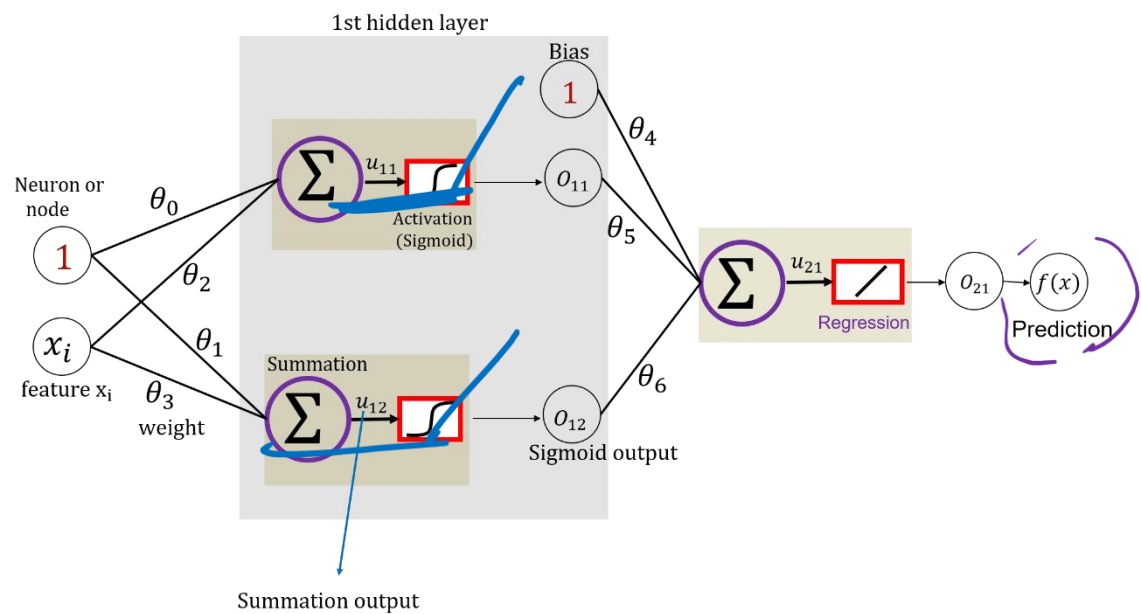
$$u_{12} = \theta_1 + \theta_3 x_i = o_{12}$$

$$u_{21} = \theta_4 + \theta_5 o_{11} + \theta_6 o_{12} = o_{21} = f(x)$$

$$f(x) = \theta_4 + \theta_5 (\theta_0 + \theta_2 x_i) + \theta_6 (\theta_1 + \theta_3 x_i)$$

$$f(x) = (\theta_4 + \theta_5 \theta_0 + \theta_6 \theta_1) + (\theta_5 \theta_2 + \theta_6 \theta_3) x_i$$

$$f(x) = \theta_0 + \theta_1 x_i$$



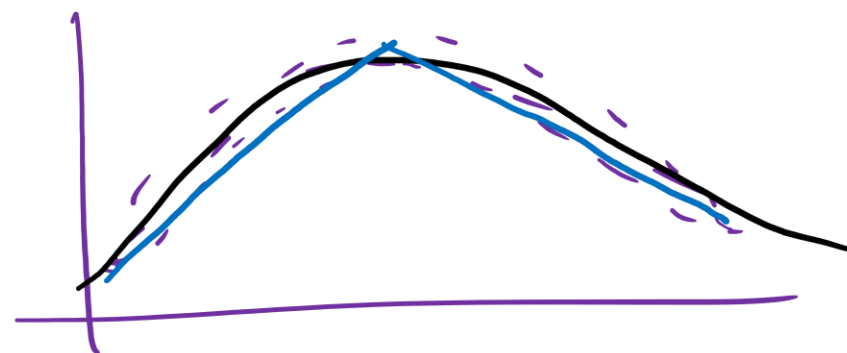
$$Q = \frac{1}{1 + \exp(-u)} * C$$

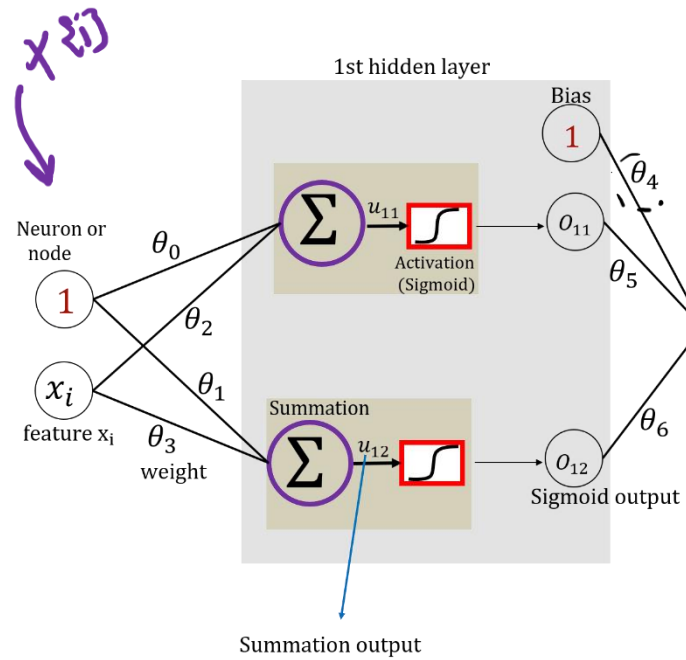
$$u_{11} = \theta_0 + \theta_2 x_i \Rightarrow Q_{11} = \frac{1}{1 + \exp(-u_{11})}$$

$$u_{12} = \theta_1 + \theta_3 x_i \Rightarrow Q_{12} = \frac{1}{1 + \exp(-u_{12})}$$

$$u_{21} = \theta_4 + \theta_5 Q_{11} + \theta_6 Q_{12} \Rightarrow Q_{21} = u_{21} = f(x)$$

$$f(x) = \underbrace{\theta_4}_{\text{vertical translation}} + \underbrace{\theta_5}_{\text{vertical stretching or squashing}} \frac{1}{1 + \exp[-(\underbrace{\theta_0}_{\text{horizontal translation}} + \underbrace{\theta_2 x_i}_{\text{horizontal stretching or squashing}})]} + \theta_6 \frac{1}{1 + \exp[-(\theta_1 + \theta_3 x_i)]}$$





② Back Propagation

$$\min E(\theta) = \frac{1}{N} \sum_{i=1}^N (y_a^{i,j} - f(x^{i,j}))^2$$

$$f(x^{i,j}) = \theta_4 + \theta_5 o_{11} + \theta_6 o_{12}$$

$$E(\theta) = \frac{1}{2} (y_a^{i,j} - f(x^{i,j}))^2$$

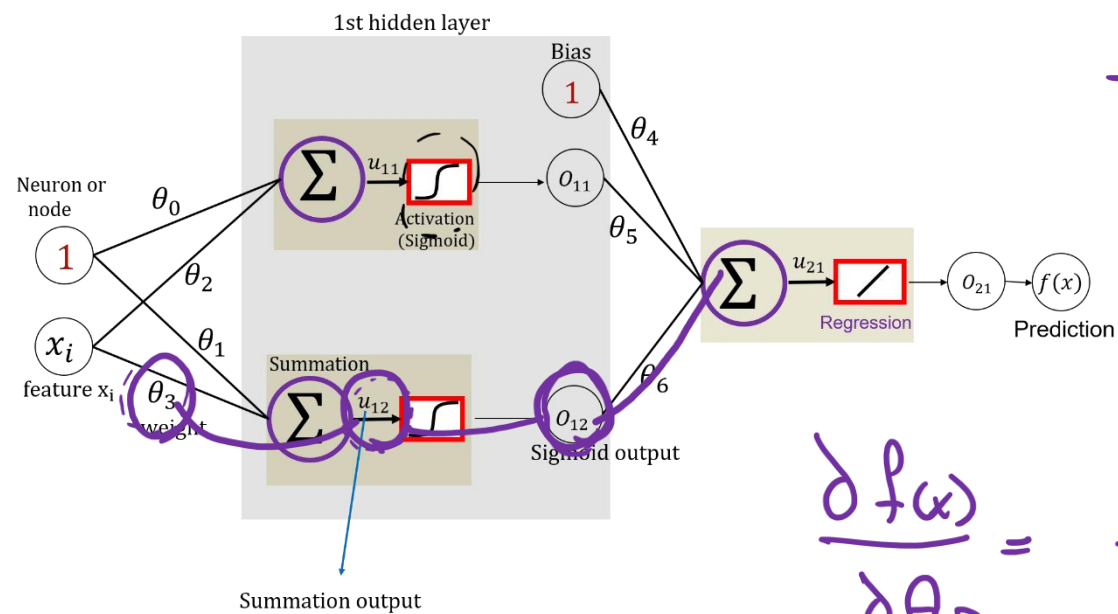
$$\nabla_{\theta} E(\theta) = \left(- (y_a^{i,j} - f(x^{i,j})) \right) \frac{\partial f(x^{i,j})}{\partial \theta} = \Delta \frac{\partial f(x^{i,j})}{\partial \theta}$$

$$\theta_0, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6 \Rightarrow \frac{\partial f(x)}{\partial \theta_0}, \frac{\partial f(x)}{\partial \theta_1}, \dots, \frac{\partial f(x)}{\partial \theta_6}$$

$$\theta_4^{\{t+1\}} \leftarrow \theta_4^{\{t\}} - \alpha \nabla_{\theta_4} E(\theta) = \theta_4^{\{t\}} - \alpha \Delta \frac{\partial f(x^{i,j})}{\partial \theta_4} = \theta_4^{\{t\}} - \alpha \Delta * 1$$

$$\theta_5^{\{t+1\}} \leftarrow \theta_5^{\{t\}} - \alpha \nabla_{\theta_5} E(\theta) = \theta_5^{\{t\}} - \alpha \Delta \frac{\partial f(x^{i,j})}{\partial \theta_5} = \theta_5^{\{t\}} - \alpha \Delta o_{11}$$

$$\theta_6^{\{t+1\}} \leftarrow \theta_6^{\{t\}} - \alpha \Delta o_{12}$$



$$f(x) = \theta_4 + \theta_5 o_{11} + \theta_6 o_{12} \quad u_{12} = \theta_1 + \theta_3 x_i$$

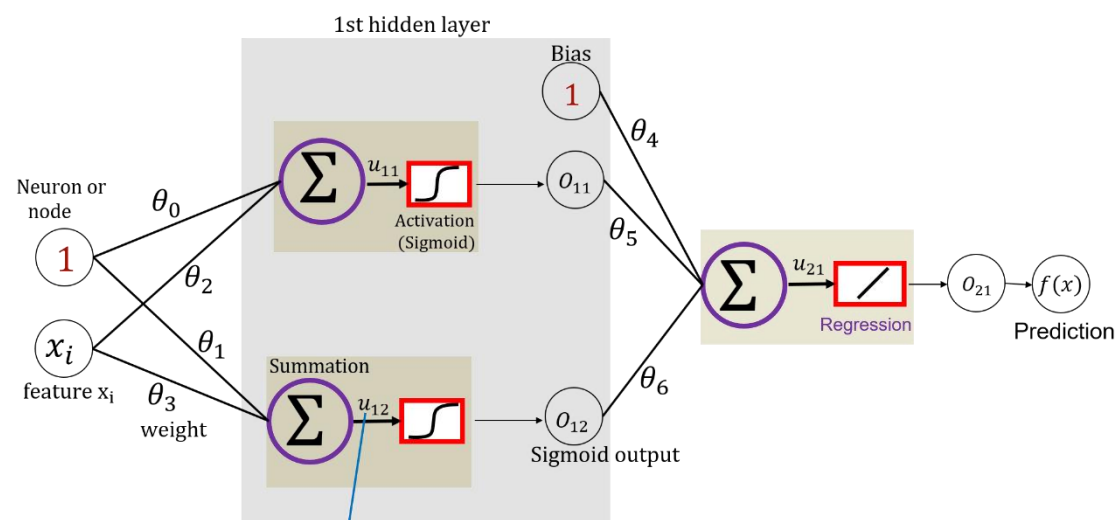
$$\theta_3^{t+1} \leftarrow \theta_3^t - \alpha \Delta \frac{\partial f(x)}{\partial \theta_3}$$

$$\frac{\partial f(x)}{\partial \theta_3} = \frac{\partial f(x)}{\partial o_{12}} \cdot \frac{\partial o_{12}}{\partial u_{12}} \cdot \frac{\partial u_{12}}{\partial \theta_3} = \theta_6 o_{12} [1 - o_{12}] x_i$$

$$O(u) = \frac{1}{1 + \exp(-u)} = (1 + \exp(-u))^{-1} \Rightarrow \frac{\partial O}{\partial u} = -1 * -1 * \exp(-u) * (1 + \exp(-u))^{-2}$$

$$\frac{\partial O}{\partial u} = \frac{(1 + \exp(-u))^{-1} - 1}{[1 + \exp(-u)]^2} = \frac{1}{1 + \exp(-u)} \left[\frac{1 + \exp(-u)}{1 + \exp(-u)} - \frac{1}{1 + \exp(-u)} \right]$$

$$\frac{\partial O}{\partial u} = O [1 - O]$$



Summation output

$$\begin{pmatrix} \theta_3 \end{pmatrix}^{t+1} \leftarrow \begin{pmatrix} \theta_3 \end{pmatrix}^t - \alpha \Delta \theta_6 o_{12} [1 - o_{12}] x_i \rightarrow 0$$

Vanishing gradient

Exploding ~

